

SentinelShield

Security Operations Center & Intrusion Detection System

Detailed Project Report

Project: SentinelShield

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1. Project Overview

SentinelShield is a lightweight cybersecurity monitoring and intrusion-detection platform developed as a practical learning project. It collects security telemetry, applies detection rules, calculates risk scores, creates alerts, stores findings in SQLite, and presents the results through a web-based Security Operations Center dashboard.

The project demonstrates the workflow of event collection, detection, risk classification, alert generation, investigation, status management and reporting.

2. Objectives

- Understand practical security-event monitoring.
- Detect common network and web attack patterns.
- Implement rule-based web attack detection.
- Detect port scanning and brute-force authentication behavior.
- Calculate 0–100 risk scores and severity classifications.
- Store security events and alerts in SQLite.
- Provide an analyst-oriented SOC dashboard.
- Generate security summaries and reports.

3. Technology Stack

Layer	Technology
Backend	Python, Flask
Database	SQLite
Network collection	Scapy
Web detection	YAML + regular expressions
Frontend	HTML, Jinja2, CSS
Environment	Windows development / isolated cybersecurity lab

4. Detection Modules

Web Attack Detection

Detects SQL injection, XSS, directory traversal and command-injection patterns using URL decoding and YAML-defined regular-expression rules.

Port Scan Detection

Tracks distinct destination ports from a source IP within a rolling time window and generates a PORT_SCAN detection when the configured threshold is crossed.

Brute Force Detection

Tracks failed authentication attempts per source IP and detects repeated failures. It also identifies a successful login following a burst of failures.

Risk Scoring

Combines event attributes into a 0–100 score and maps the result to LOW, MEDIUM, HIGH or CRITICAL.

Alert Lifecycle

Supports NEW → ACKNOWLEDGED → INVESTIGATING → RESOLVED status transitions.

5. SOC Dashboard

The dashboard provides alert counts, severity distribution, threat activity, event timelines, detection-engine status and recent security alerts. The Alerts page supports filtering and incident management. Alert Detail provides risk, evidence, source and destination information, recommended action, related events and status controls. Network, Web Attacks and Reports provide specialized views.

6. Database and Investigation Data

At a recorded project checkpoint, the SQLite database contained 57 events and 20 alerts. Alerts included timestamp, event type, source IP, destination IP, severity, confidence, risk score, description, evidence, recommended action, status, engine and related event IDs.

7. Detection Test Results

Test	Result
SQL Injection	Detected
XSS	Detected
Directory Traversal	Detected
Command Injection	Detected
Port Scan	Detected
Brute Force	Detected
Successful Brute Force	Detected

8. Practical Workflow

- Start the Flask application and open the SOC dashboard.
- Generate or replay controlled laboratory events.
- Run detection modules against the events.
- Create and store alerts in SQLite.
- Review alerts using severity and status filters.
- Open an alert investigation view and inspect evidence and related events.

- Move an incident through its lifecycle.
- Review Network and Web Attack telemetry.
- Use the Reports page to summarize findings.

9. Security and Lab Safety

The project is intended for authorized and isolated cybersecurity laboratory environments. Network packet capture should only be enabled on an interface belonging to the controlled lab. The web detector analyzes suspicious patterns and does not construct or execute attack payloads.

10. Limitations

- Pattern-based rules can produce false positives and false negatives.
- Threat-intelligence integration is not part of the current core implementation.
- SQLite is appropriate for a student/lab deployment but not for high-volume production telemetry.
- The project is educational and should not be represented as a replacement for a production SIEM or WAF.
- Live packet capture depends on interface configuration and required operating-system privileges.

11. Future Enhancements

- Role-based access control and analyst authentication.
- Persistent analyst investigation notes.
- Suricata EVE JSON integration.
- Threat-intelligence enrichment.
- Configurable detector thresholds.
- PDF/CSV report export.
- Historical security analytics.
- Production-grade telemetry storage.

12. Submission Evidence

Item	Submission requirement
GitHub repository	Public HTTPS repository URL
Detailed report	Public HTTPS link to this PDF
Deployed project	Live HTTPS URL
Feedback video	Unlisted/public video URL
Screenshots	Dashboard, Alerts, Alert Detail, Network, Web Attacks, Reports
Testing evidence	Complete detection test output

Replace the submission form fields with the actual URLs after publishing the project.